TEXORA · SESSIONAL NOTES · LAB EXPERIMENT SHEET

Search, Generate, Manipulate Data Using MS Office/Open Office

The spreadsheet as a working tool: entering a clean table of data, letting formulas — never re-typed arithmetic — do the calculations, and then searching, sorting and filtering the table until it answers questions.

CSE-2106 · FUNDAMENTALS OF COMPUTING SESSIONAL · EXPERIMENT 1 · LEVEL 2 TERM 1

1. Aim

To create a data table in a spreadsheet (MS Excel or LibreOffice Calc — the operations are the same), generate calculated columns with formulas and built-in functions, and manipulate the data by searching, replacing, sorting and filtering.

2. Concepts you need

- **Workbook, worksheet, cell:** a file is a workbook of one or more worksheets; every box on the grid is a **cell**, addressed by column letter + row number (B2 = column B, row 2).
- **Data types:** text (labels), numbers (values you can calculate with) and formulas. A formula always begins with =.
- **References:** a *relative* reference (B2) shifts when the formula is copied; an *absolute* reference (B2) stays fixed. Pressing F4 toggles the forms.
- **Core functions:** `SUM`, `AVERAGE`, `MAX`, `MIN`, `COUNT`, `COUNTA`, `COUNTIF`, `IF`. A range is written B2:D2 (across a row) or B2:B6 (down a column).
- **Manipulation tools:** Find & Replace (Ctrl+H) to *search* and correct entries; Sort to reorder rows by a key; Filter to display only rows meeting a condition — the data is hidden, not deleted.

fx =SUM(B2:D2) FORMULA BAR (cell E2 selected)

A · NAME	B · PHY	C · CHE	D · MATH	E · TOTAL	F · AVERAGE
Rahim	78	65	82	225	75.00
Karim	56	71	60	187	62.33
Selina	90	88	95	273	91.00
Arif	45	52	48	145	48.33

THE FORMULA LIVES IN E2 — THE NUMBER 225 IS ONLY ITS RESULT

Figure 1 — The marks worksheet of the worked example. Totals and averages are formulas, so correcting any mark recalculates the whole row at once.

3. Lab procedure

1. Open the spreadsheet. In row 1 type the headings: Name, Physics, Chemistry, Mathematics, Total, Average.
2. Enter the five students' marks given in the worked example (rows 2–6). Keep one fact in one cell — never type "78 marks" into a number cell.
3. In E2 enter `=SUM(B2:D2)`; in F2 enter `=AVERAGE(B2:D2)`. Select E2:F2 and fill down to row 6.
4. Below the table compute the class statistics: highest total `=MAX(E2:E6)`, lowest `=MIN(E2:E6)`, and mean of the totals `=AVERAGE(E2:E6)`.
5. **Search:** use Find (Ctrl+F) to locate "Selina". **Replace:** a name was entered as "Arif" in one row and "Aref" in another — use Find & Replace to make all entries consistent. This is how real data is cleaned.
6. **Sort:** select the whole table (including headings), sort by Total, largest to smallest. The merit order appears; the rows move as complete records — never sort one column alone, or names and marks part company.
7. **Filter:** turn on AutoFilter and set Total ≥ 200. Only the qualifying students show. Clear the filter and confirm all five rows return.
8. **Generate:** add a column G, "Result", with `=IF(F2>=40,"Pass","Fail")` filled down — a new column generated entirely by a rule.
9. Save the file in both native format and, from a copy, export a PDF — the two deliverable forms of office work.

4. Worked example — resulting table

NAME	PHYSICS	CHEMISTRY	MATHEMATICS	TOTAL	AVERAGE
Rahim	78	65	82	225	75.00
Karim	56	71	60	187	62.33
Selina	90	88	95	273	91.00
Arif	45	52	48	145	48.33
Mina	67	74	71	212	70.67

VERIFY THE SHEET BY HAND ONCE

Totals: 78+65+82 = 225; 56+71+60 = 187; 90+88+95 = 273; 45+52+48 = 145; 67+74+71 = 212. Averages are totals ÷ 3 (225÷3 = 75.00; 187÷3 = 62.33…). Class figures: MAX = 273, MIN = 145, mean of totals = (225+187+273+145+212) ÷ 5 = 1042 ÷ 5 = **208.4**. Sorted descending by Total the order is Selina, Rahim, Mina, Karim, Arif; the filter Total ≥ 200 shows Selina, Rahim and Mina. If your sheet disagrees with any of these, a formula — not the arithmetic — is wrong: click the cell and read its formula in the formula bar.

5. Practice tasks

1. Add a column "Percentage" (total out of 300). State the formula for the first data row and the five values it should produce.
2. Write the formula that counts how many students scored 60 or more in Chemistry (column C), and state its value for the example data.
3. A mark for Mina's Mathematics was typed as 17 instead of 71. Without re-typing any formula, give the corrected Total and Average for Mina after you fix the entry, and explain why they update themselves.

Answer key

1. In G2: `=E2/300*100`. Values: Rahim 75.00, Karim 62.33, Selina 91.00, Arif 48.33, Mina 70.67 (the same as the averages, because the average of three marks out of 100 *is* the percentage — a useful check).
2. `=COUNTIF(C2:C6,">=60")` → Chemistry marks are 65, 71, 88, 52, 74 → four are ≥ 60 → **4**.
3. Correcting D6 from 17 to 71 changes only that input cell. Mina's Total becomes 67+74+71 = **212** and Average **70.67**, because E6 and F6 hold formulas that reference D6 — a spreadsheet recalculates every dependent formula whenever an input changes. That is the whole reason results are never typed in by hand.